

Computer Vision

1. Gradient magnitude and Direction

Given

$$G_x = 24 \quad G_y = 18$$

Gradient magnitude

$$m = \sqrt{G_x^2 + G_y^2}$$

$$m = \sqrt{(24)^2 + (18)^2} = \sqrt{576 + 324} = \sqrt{900} = 30$$

Gradient direction

$$m = 30$$

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{18}{24} \right) \Rightarrow \tan^{-1} (0.75) \approx 36.87^\circ$$

$$\theta = 36.87^\circ$$

2. Non maximum . Suppression

Pixel	P1	P2	P3	P4	P5
magnitude	25	50	90	60	30

keep the pixel value if it strictly greater than both its neighbors otherwise suppress it to 0.

$$P2 (50) = 50 \rightarrow 0 \text{ as it depends on } \rightarrow 0.$$

$$P_3(90) = P_2(60) P_4(60).$$

~~50~~ $90 > 50, 90 > 60.$
 $\rightarrow 90.$

$$P_4(60) = P_3(90) P_5(30).$$

~~60~~ $90 > 50, 90 > 30.$
 $60 < 90 \rightarrow 0.$

$P_3 = 90$
$P_2, P_4 = 0.$

Pixel Classification by Thresholding.

High Threshold = 100

Low Threshold = 50

magnitude $\geq 100 \rightarrow$ Strong Edge

$50 \leq \text{mag} < 100 \rightarrow$ Weak Edge

$\text{mag} < 50 \rightarrow$ Non Edge

$P_1(135) \geq 100 \rightarrow$ Strong Edge

$P_2(90) 50 < 90 < 100 \rightarrow$ Weak Edge

$P_3(45) < 50 \rightarrow$ Non Edge

$P_4(120) > 100 \rightarrow$ Strong Edge

$P_5(70) 50 < 70 < 100 \rightarrow$ Weak Edge

$P_6(20) < 50 \rightarrow$ Non Edge

4.

Crises

P1 130

P2 95

P3 40

P4 115

P5 65

P6 25

P7 105

High Threshold = 100

Low Threshold = 50

P2 is connected to P1

P5 is not connected to any strong edge

Strong Edges ≥ 100 condition

P1 = 130 P4 = 115 P7 = 105

Strong Edges: P1, P4, P7

Weak Edges 50 & 99

P2 = 95

P5 = 65

Weak Edges P2, P5

Non Edges

50

P3 = 40 P6 = 25

Non Edges P3, P6.

Final Edge pairs: P1, P2, P4, P7.